Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/26/2008
PI Name:	Winter Sederoff, Heike Ph.D.		
Project Title:	Transcriptional and Translational Regulation of Gravitropism in the Maize Pulvinus		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davies, Eric (Dept. Botany, North Carolina State University) Perera, Imara (Dept. Botany, North Carolina State University)		
Grant/Contract No.:	NNA04CC56G		
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Task Description:	Gravitropic bending in cereal grass stems is brought about by differential growth of the pulvinus, a specialized tissue which senses and responds to gravity. The maize pulvinus is an ideal model system to study plant gravitropism because the gravistimulated pulvinus can be separated into tissues with different rates of cell elongation (upper vs lower halves) and the gravity signal transduction cascade has been biochemically characterized. However, little is known about the regulation of gene expression networks during gravitropism and the mechanisms involved in transcriptional and translational control. Differences in transcript levels and the association of calmodulin and calreticulin mRNA with polyribosomes correlates positively with differential cell elongation during the gravitropic response of the maize pulvinus (Heilmann et al. 2001). Our hypothesis is that translational control of selective transcripts is an essential regulatory event in plant gravitropic signal transduction and response. Specific Aims: 1. To identify genes specifically regulated at the transcriptional and translational level during a time-course of gravitropism by microarray analysis of total vs polyribosomal RNA from both upper and lower pulvini halves. 2. To select transcripts, assess their functional role in gravitropism and determine their cellular localization. 3. To understand the genetic requirements of the gravitropic response by comparing transcript profiles of the gravistimulated maize pulvinus with data available from other plant systems. This work will provide the first comprehensive overview of genetic regulation at the transcriptional and translational level of gravitropism in a major crop species. This information will be invaluable in generating tools to manipulate a plant's gravitropic response for agricultural benefits on Earth and in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Maize is the major cereal crop in the USA and in the World, but yield suffers when the plant lodges, (i.e., it is blown over by wind, hail, or rain), and rotting ensues thereby damaging the crop. One way to mitigate this loss of yield is making the stem stronger. Extra lignification would make the stems less biodegradable. Another approach is to understand the highly specialized system in maize, which enables the plant to reorient itself in response to gravity. This proposal focuses on understanding this reorientation mechanism with the ultimate aim of making maize more rapidly gravi-responsive. We anticipate that this will ultimately reduce the time the maize plant takes to elevate the ears away for the ground, and thus salvaging crop yield. Other applications of this work could result in improved quality of cut flowers (maintenance of straight stems during long distance shipping). Understanding the response of plants to gravity also has practical implications for the wood and paper products industry. Reaction wood formed in response to gravity, particularly compression wood in conifers degrades wood quality and increases toxic products during chemical pulping. Understanding the gravitational response at a molecular level for trees could lead to improved wood products and reduced environmental impact. This research will determine the mechanisms by which the maize pulvinus senses and responds to gravity. In addition, it will provide a fundamental understanding of basic plant processes, especially at the molecular and developmental levels. A knowledge of how plants respond to gravity and other environmental conditions will improve our understanding of how they will grow in various space conditions under less than unit gravity (e.g. Earth orbit, Mars) and how their growth can be modified to maximize yields on Earth.
Task Progress:	Maize responds to gravitational changes by differential growth of the pulvinus. This specialized tissue is an ideal model to study the underlying processes that are involved in plant gravitropism. The signal transduction cascade involved in plant gravitropism has been studied at the biochemical level, but the underlying molecular processes are not well understood. Using microarray technology, we investigated the transcriptional and translational regulation of gene expression in response to gravity stimulation in maize pulvini during the first hour after reorientation. Six week old maize plants were gravity stimulated by 90º reorientation. Upper (slow elongation) and lower (fast elongation) halves of the most gravity competent pulvini were harvested over a time course ranging from 2 minutes up to one hour. For each time point, total mRNA and polyribosome-associated mRNA were purified and the transcript profiles analyzed using Affymetrix GeneChip® Maize Genome Arrays. A total of 13584 transcripts showed levels of expression that were above background signal. Changes in the relative abundance of transcripts were seen the total mRNA fractions were compared with the mRNA we isolated from the polyribosomes. We found transcripts that show higher transcript abundance in the total mRNA than in the polysomal mRNA fraction. This suggests that these transcripts might be regulated at the translational level. We also found that some transcripts changed in transcript abundance over time even though similar expression profiles were seen when total mRNA and polysomal mRNA were compared. This result proposes that for these transcripts regulation most likely occur at the transcriptional level. During this time course, no significant quantitative or temporal differences in transcription or translation of genes were found be tween the "upper" and "lower" halves of the maize pulvinus, indicating that differences in the kinetics of bending do either not occur during the first hour after stimulation or are not at the level of gene expression. To identify conserved mechanisms in the gravity induced gene expression in plants, we compared the transcript profiles from gravity stimulated maize pulvini with transcript profiles from gravity stimulated Arabidopsis root apices (Kimbrough et al., 2004). Initial comparison identified four orthologs between corn pulvini and Arabidopsis root apices that show the same temporal and quantitative changes in transcript abundance after gravity stimulation. Functional analysis for one of these genes has confirmed its requirement in the bending response during gravitropism.
Bibliography Type:	Description: (Last Updated: 03/24/2008)
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Awards	Hammond RL, Myburg H, Perera IY, Brown CS, Winter Sederoff H. "Second Place Undergraduate Poster Competition for “Is Transcriptional Regulation of Gravitropism Conserved Between Dicots and Monocots?“ Annual Meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 2006. " Nov-2006
Awards	Yester JW, Kimbrough JM, Brown CS, Winter Sederoff H. "First Place, Undergraduate Poster Competition, “Signal transduction elements involved in the regulation of gravity specific gene expression in Arabidopsis roots,“ ASGSB annual meeting, Reno, Nevada, November 2005." Nov-2005
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